

Window Function Queries (SQL) => * Grouping of data alongside individual references =>

For example : Display Individual/Highest Students Individual Marks Highest

Students	Individual Marks	Highest
S1	80	96
S2	85	96
S3	72	96
S4	65	96
S5	96	96

BCNF => Boyce - Codd Normal Form 1NF, 2NF, 3NF -> This one is a slightly stricter version of 3NF.

Rule => Every determinant must be a candidate key.

{ If Column A determines Column B, then A must be a key that uniquely identifies a row. No exception }

Student	Subject	Teacher
Preethi	Maths *	Mr. Roy
Pratham	Science	Ms. Patel
Punith	Maths *	Mr. Roy

Here => Teacher -> Subject (knowing teacher tells you the subject). But Teacher is not a key - it's just a regular column. This violates BCNF. If Mr. Roy switches to a different subject (Physics) you have to update multiple rows.

The BCNF Fix => Just Split it.

Teacher	Subject	Student	Teacher
Mr. Roy	Maths	Preethi	Mr. Roy
Ms. Patel	Science	Pratham	Ms. Patel
		Punith	Mr. Roy

* Now Teacher is a key in its own table.

- This is the solution using BCNF.

Sample Mini-Project =>

Instagram-Like DB System

Objective =>

- Build a mini-Instagram backend
- * users create accounts
- * users create posts
- * users like posts
- * users follow each other

Learning: (1) Relationships (ER) (2) Joins (3) Aggregates (4) How real world db works.

File Handling in Java =>

It is the process of creating, deleting, reading, writing of data from or to files and folders using a programming language [Java]

Most Interviews => CRUD Operations

- 1 Create Folder
- 2 Delete Folder
- 3 Create File
- 4 Delete File
- 5 Read data from file
- 6 Write data into file
- { text files (.txt) }

Latest (Java) Interviews => 80-90% J 30% C++

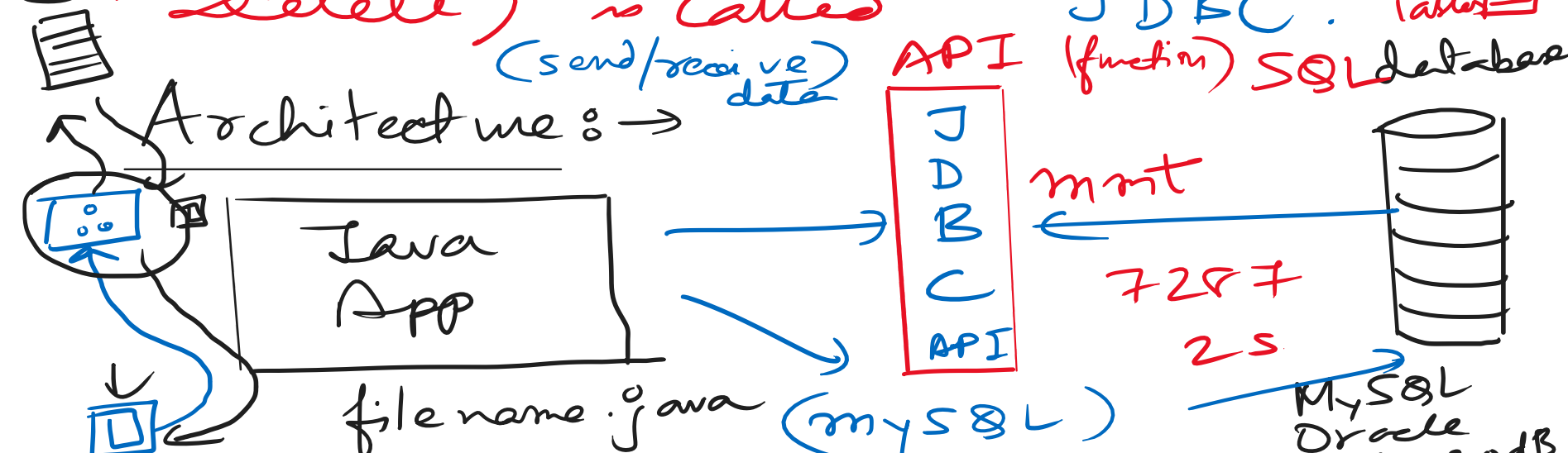
Oracle, IBM, TCS, Wipro, HP, HCL, Amazon, (Pure Storage, Mercedes) -> USSSIETW, M

(RVCE)

(Java 1.8) => 1.9 -> (Streams) {:::} Mapping Reducing Filtering Collecting (Lambda Expressions)

JDBC -> Java Database Connectivity

The process of connecting the Java application to backend to perform CRUD (Create, Read, Update, delete) is called JDBC. Tailor



Components of JDBC => DELIMITER

- DriverManager (Class): Manages & establishes connection b/w Java Application & Database.
- Driver (Interface): Decides which db we are connecting to.
- Connection (Interface): Represents a connection with specific database.
- Statement (Interface): Used to execute simple SQL queries without parameters.
- PreparedStatement (Interface): Used to execute complex SQL queries with parameters.
- CallableStatement (Interface): Used to execute stored procedures in SQL.
- ResultSet (Interface): Used to store the result of a SQL query.
- SQLException (Class): Handles database-related exceptions and errors.

The library containing all the

JDBC Components is the package [java.sql] (JAR) JAVA APPLICATION ARCHIVE

* Important -> MySQL WB

*** [mysql-connector-jar-8.0.31] db